

Nested Queries & Aggregations

24L-0583 Lab...d Raza (57)

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```
--  
-- PART 1: Nested Queries & Aggregations  
--  
-- 1. Patients infected with 'Dartu Virus' using subquery  
SELECT Name  
FROM Patients  
WHERE PatientID IN (  
    SELECT PatientID  
    FROM Infections  
    WHERE GermID = (  
        SELECT GermID  
        FROM Germs  
        WHERE Name = 'Dartu Virus'  
    )  
);  
GO  
  
--  
-- PART 2: Views  
--  
-- 1. View for high impact germs (ImpactLevel >= 8)  
CREATE VIEW vw_HighImpactGerms AS  
SELECT *  
FROM Germs  
WHERE ImpactLevel >= 8;  
GO
```

90 %

No issues found

Ln: 6, Ch: 1 | (207 chars, 12 lines) SPC CRLF UTF-8

Results Messages

	Name
1	Sara Ahmed

Views

24L-0583 Lab...d Raza (57)

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```
FROM Germs  
WHERE Name = 'Dartu Virus'  
);  
GO  
  
--  
-- PART 2: Views  
--  
-- 1. View for high impact germs (ImpactLevel >= 8)  
CREATE VIEW vw_HighImpactGerms AS  
SELECT *  
FROM Germs  
WHERE ImpactLevel >= 8;  
GO  
  
-- Test it  
SELECT * FROM vw_HighImpactGerms;  
GO  
  
--  
-- PART 3: Stored Procedures  
--
```

90 %

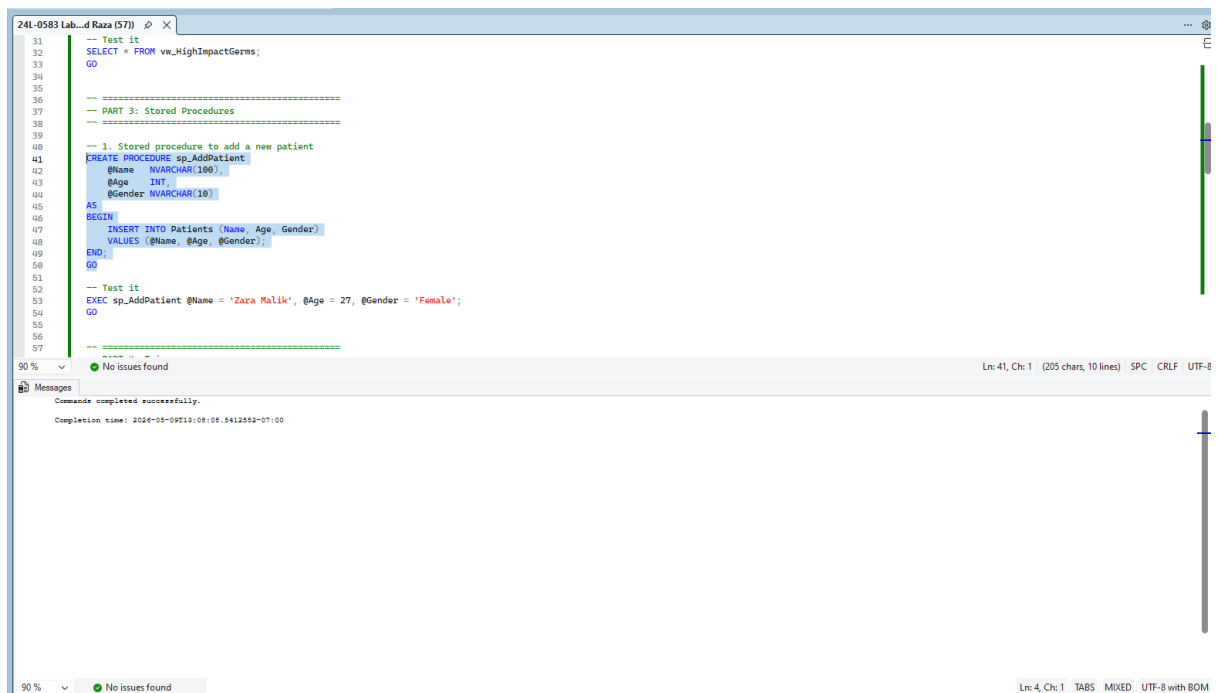
No issues found

Ln: 24, Ch: 1 | (190 chars, 10 lines) SPC CRLF UTF-8

Results Messages

	GermID	Name	Role	ImpactLevel	DiscoveredDate
1	1	Dartu Virus	Airborne	9	2024-01-15
2	3	Dart Parasite	Waterborne	8	2024-02-20
3	6	Jerm Attack	Airborne	8	2024-02-10
4	7	Bacteria Khan	Contact-Based	9	2023-10-22
5	10	Contagion King	Airborne	10	2023-09-15

Stored Procedures



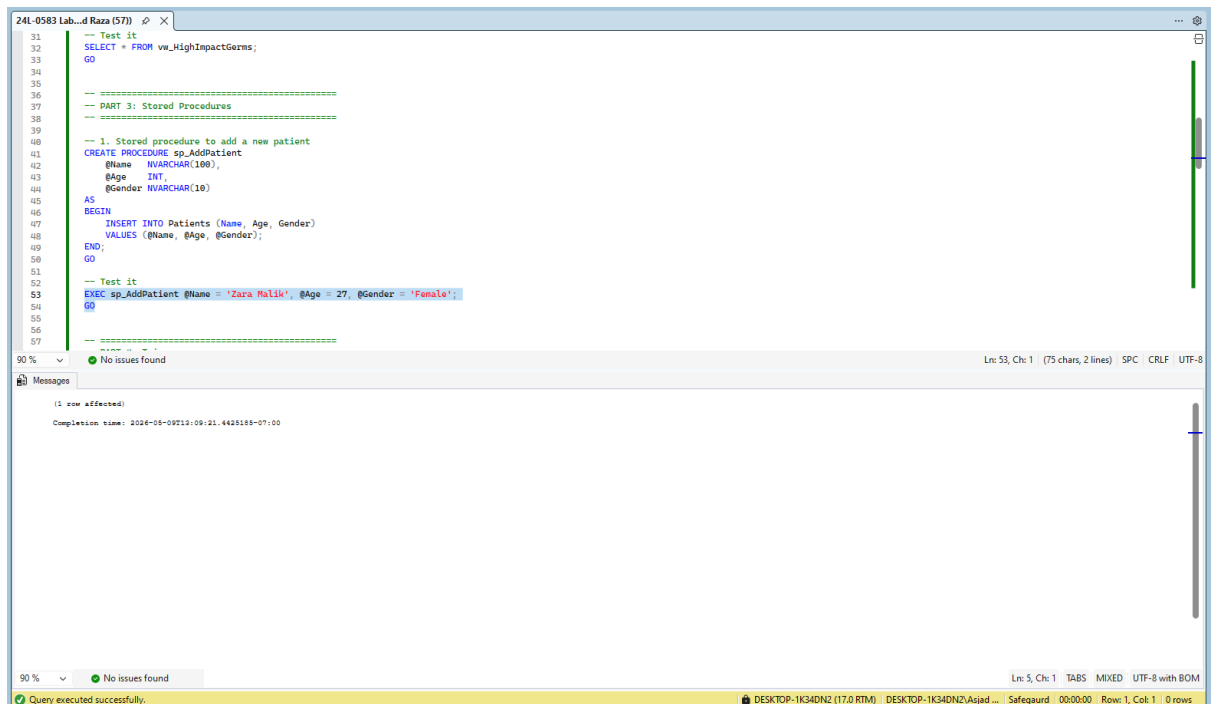
The screenshot shows a SQL Server Enterprise Manager window titled "241-0583 Lab...d Raza (57)". The main pane displays a SQL script with the following content:

```
31 -- Test it
32 SELECT * FROM vw_HighImpactGerms;
33 GO
34
35
36
37 -- PART 3: Stored Procedures
38
39
40 -- 1. Stored procedure to add a new patient
41 CREATE PROCEDURE sp_AddPatient
42     @Name NVARCHAR(100),
43     @Age INT,
44     @Gender NVARCHAR(10)
45 AS
46 BEGIN
47     INSERT INTO Patients (Name, Age, Gender)
48     VALUES (@Name, @Age, @Gender);
49 END;
50 GO
51
52 -- Test it
53 EXEC sp_AddPatient @Name = 'Zara Malik', @Age = 27, @Gender = 'Female';
54 GO
55
56
57
```

The status bar at the bottom indicates "Ln: 41, Ch: 1 (205 chars, 10 lines) SPC CRLF UTF-8". The Messages pane shows the following message:

Command completed successfully.
Completion time: 2024-05-09T19:08:08.5412552-07:00

Test



The screenshot shows the same SQL Server Enterprise Manager window. The main pane displays the same SQL script as in the previous screenshot. The Messages pane now shows the following message:

(1 row affected)
Completion time: 2024-05-09T19:09:21.4425185-07:00

The status bar at the bottom indicates "Ln: 53, Ch: 1 (75 chars, 2 lines) SPC CRLF UTF-8". The bottom status bar shows "Query executed successfully." and "DESKTOP-1K34DN2 (17.0 RTM) | DESKTOP-1K34DN2\Ajad ... Safeguard 00:00:00 Row: 1, Col: 1 | 0 rows".

Triggers

```
24L-0583 Lab...d Raza (57)
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56
57 -- =====
58 -- PART 4: Triggers
59 -- =====
60
61 -- 1. Audit trigger -- fires AFTER INSERT on Prescriptions
62 CREATE TRIGGER tr_AuditPrescriptions
63 ON Prescriptions
64 AFTER INSERT
65 AS
66 BEGIN
67     INSERT INTO Prescriptions_History (PrescriptionID, PatientID, PrescriptionDate, ValidFrom)
68     SELECT
69         PrescriptionID,
70         PatientID,
71         PrescriptionDate,
72         GETDATE()
73     FROM inserted;
74 END;
75 GO
76
77 -- Test it (trigger fires automatically)
78 INSERT INTO Prescriptions (PatientID, PrescriptionDate)
79 VALUES (1, GETDATE());
80 GO
81
```

90 % No issues found Ln: 62, Ch: 1 (308 chars, 14 lines) SPC CRLF U

Messages

Command completed successfully.

Completion time: 2026-05-09T13:10:11.0297711+07:00

90 % No issues found Ln: 4, Ch: 1 TABS MIXED UTF-8 with B

```
24L-0583 Lab...d Raza (57)
62 CREATE TRIGGER tr_AuditPrescriptions
63 ON Prescriptions
64 AFTER INSERT
65 AS
66 BEGIN
67     INSERT INTO Prescriptions_History (PrescriptionID, PatientID, PrescriptionDate, ValidFrom)
68     SELECT
69         PrescriptionID,
70         PatientID,
71         PrescriptionDate,
72         GETDATE()
73     FROM inserted;
74 END;
75 GO
76
77 -- Test it (trigger fires automatically)
78 SELECT * FROM Prescriptions_History;
79 GO
80
81 INSERT INTO Prescriptions (PatientID, PrescriptionDate)
82 VALUES (1, GETDATE());
83 GO
84
85 SELECT * FROM Prescriptions_History;
86 GO
87
88
```

90 % No issues found Ln: 78, Ch: 1 (171 chars, 9 lines) SPC CRLF UTF-8

Results

	PrescriptionID	PatientID	PrescriptionDate	ValidFrom	ValidTo
1	6	1	2026-05-09 13:10:48.2233333	2026-05-09 13:10:48.2266667	NULL

Query executed successfully.

DESKTOP-1K34DN2 (17.0 RTM) DESKTOP-1K34DN2\Asjed ... Safeguard 00:00:00 Row: 1, Col: 1 1 rows

Transactions

241-0583 Lab...d Raza (57f) X

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```
--  
-- PART 5: Transactions  
--  
-- 1. Insert a Prescription + 2 Prescription_Remedies atomically  
BEGIN TRANSACTION;  
BEGIN TRY  
    -- Step 1: Insert the prescription  
    INSERT INTO Prescriptions (PatientID, PrescriptionDate)  
    VALUES (2, GETDATE());  
    -- Grab the new PrescriptionID  
    DECLARE @NewPrescriptionID INT = SCOPE_IDENTITY();  
    -- Step 2: Insert first remedy  
    INSERT INTO Prescription_Remedies (PrescriptionID, RemedyID, Dosage)  
    VALUES (@NewPrescriptionID, 1, 'Once daily');  
    -- Step 3: Insert second remedy  
    INSERT INTO Prescription_Remedies (PrescriptionID, RemedyID, Dosage)  
    VALUES (@NewPrescriptionID, 2, 'Twice daily');  
    COMMIT TRANSACTION;  
    PRINT 'Transaction committed successfully.';  
END TRY  
BEGIN CATCH  
    ROLLBACK TRANSACTION;  
    PRINT 'Error occurred: transaction rolled back.';  
    PRINT ERROR_MESSAGE();  
END CATCH;  
GO
```

90 % No issues found Ln: 94, Ch: 1 (814 chars, 30 lines) SPC CRLF UTF-

Messages

(1 row affected)
(1 row affected)
(1 row affected)
(1 row affected)
Transaction committed successfully.
Completion time: 2026-08-09T10:11:17.0210880+07:00

90 % No issues found Ln: 12, Ch: 1 TABS MIXED UTF-8 with BOM